Nithusikan Thasavaran

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Profile

A fresh Computer Engineering graduate passionate about AI, Machine Learning, and Data Science. Experienced in NLP research, applied deep learning, software engineering and web scraping. Proven ability to lead technical projects and deliver scalable solutions. Eager to apply data-driven approaches to real-world challenges.

Education

University of Peradeniya, Sri Lanka **2021 March - present**
BScEngHons specializing in Computer Engineering *3.50/4.00 CGPA*

Jaffna Hindu College, Sri Lanka **2011 - 2019**
G.C.E Advanced Level Examination — Results - 3A District Rank - 46 Island Rank - 562 *2.1369 Z-Score*

Experience

LIRNEasia Colombo, Sri Lanka
Data Science Intern July 2024 —- January 2025

- Worked as a Lead Data Science Researcher in a workstream of a key NLP project, building models to predict and analyze tweet virality, and optimized for a balanced trade-off between precision and recall.
- Engineered features from tweet content and metadata, applying text vectorization and sentiment analysis techniques (FastText, Word2Vec, TF-IDF, VADER) to improve model performance.
- Developed a modular command line tool that automated data cleaning, feature engineering, model training, and evaluation, enhancing reproducibility and scalability across experiments.
- The hashtag networks were analyzed using clustering and graph-based techniques (Louvain, Spectral, NetworkX) and the patterns were visualized using Plotly and Neo4j, uncovering thematic social media trends.
- Built web scraping pipelines (Scrapy, Selenium, Playwright, BeautifulSoup, lxml) to collect and preprocess high-quality datasets for downstream analysis.

Technical Skills

Languages: Python, Java, C, JavaScript, SQL, Verilog HDL. HTML, CSS
Frame Works: LangChain, SpringBoot, React.js, Electron.js,FastAPI , Bootstrap
Databases: MySQL, MongoDB, Neo4j
AWS Services: AWS-EC2, AWS-S3, AWS-Lambda, AWS-IAM, AWS-SageMaker, AWS-IOT Core
Tools & Libraries: Pandas,Scikit-Learn, PySpark, TensorFlow, PyTorch, Hugging Face, PowerBI, Postman

Projects

1. A Cross-Layer Cyberattack Detection Framework: Group — 2025 Mar - 2025 Aug

- Our final year project which is expected to develop an end to end pipeline to effectively identify known and unknown cyber attacks by analyzing Network traffic and flagging the potentially affected components through Application layer logs.
- **Contributions:**
  - Engineered the solution architecture by coordinating with my team mates and referring to existing research papers.
  - Handled the entire network layer analysis and trained the necessary ML models to meet our requirements.
  - Developed the attack detection pipeline for network layer traffic.
- **Tech Stack: Python, Pandas, Scikit- learn, Matplotlib, Seaborn, Numpy, PyTorch**

2. ML Based Forecasting of Inpatient Bed Demand: Group — 2023 Nov - 2024 Apr

- A ML powered application is being developed to forecast inpatient bed demand, aiming to enhance hospital accessibility, reduce wait times, improve quality of care, and boost satisfaction among both patients, employees and management.
- **Contributions:**
  * Played a key role in training various models and evaluating the accuracy of the forecast.
  * Led the data analysis phase using PowerBI during preprocessing.
- **Tech Stack: Python, Pandas, Scikit- learn, Matplotlib, Seaborn, Numpy, PowerBI, TensorFlow**

3. Vehicle Number Plate Recognition System: Individual — 2024 Apr - Present

- Developed an advanced application to detect vehicles and automatically recognize number plates in real-time.
- Designed and implemented a comprehensive computer vision pipeline for Automatic Number Plate Recognition (ANPR):
  * Used the YOLOv8 deep learning model for precise real-time vehicle detection.
  * Integrated EasyOCR library for high-accuracy character recognition from detected license plates.
- **Tech Stack: Python, (YOLOv8, EasyOCR), OpenCV**

4. Customer Support Chatbot: Individual — 2024 Apr - Present

- Developing an intelligent customer support chatbot by fine-tuning Llama-3 with Retrieval-Augmented Generation (RAG) using the Bitext-customer-support-llm-chatbot-training-dataset from **Hugging Face** to deliver exceptional customer service.
- **Contributions:**
  * Fine-tuned the Llama-3 (4 - bit quantized) model using the documentation and Unsloth repository.
  * Implemented the RAG pipeline using ChromaDB and LangChain.
- **Tech Stack: Python, Llama-3, PyTorch, ChromaDB, LangChain**.

5. Automated Hydroponics System: Group — 2023 Nov - 2024 Jan

- An IoT-based hydroponics system integrated with web and mobile applications for real-time monitoring, control of system conditions, and resource management.
- **Contributions:**
  * Implemented the AWS connectivity between the backend application and physical IoT system using Spring Boot and AWS IoT Core with **MQTT** protocol.
  * Developed the backend application for both mobile and web application using **JWT** token for protection.
- **Tech Stack: Spring Boot(Java), React.js, AWS IoT Core, MySQL**

6. Engineering Education Unit: Group — 2023 AUG - OCT

- A web application provides a dynamic environment to support education, foster connections between staffs and learners, and offer materials worldwide. **Deployed in department server** and actively serving the users.
- **Contributions:**
  * Engineered the creation of the authentication and authorization system utilizing Spring Security, JWT token authentication, and secured routing.
  * Designed the fronted application using figma for ensuring the easiness and faster develpment.
  * Undertook the development of both frontend and backend components.
- **Tech Stack: Spring Boot(Java), React.js, MySQL, Figma**

Certifications

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning
- Introduction to Linux

Achievements

- ACES - Hackfest - 2023 **Winners** of the sustainable development category
- ACES - Coders V10.0 **Final round participants**

References

Prof. Roshan G. Ragel, Dept. of Computer Engineering, University of Peradeniya,roshanr@eng.pdn.ac.lk
Dr. Isuru Nawinne, Dept. of Computer Engineering, University of Peradeniya,isurunawinne@eng.pdn.ac.lk
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